

ENTHUSIAST COURSE [PHASE-ALL]

TARGET : PRE-MEDICAL 2020

Test Type : Full Syllabus

Test # 02

Test Pattern : NEET-UG

TEST DATE : 25 - 04 - 2020

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	1	4	2	3	1	3	2	4	2	3	3	3	4	3	3	2	4	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	1	3	1	3	3	2	1	3	2	2	1	2	3	2	2	1	1	3	3
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	3	1	4	4	3	3	3	1	4	1	2	2	1	4	4	2	4	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Ans.	3	1	4	2	3	2	3	4	1	2	3	4	3	3	3	4	2	3	2	2
Que.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Ans.	3	3	4	3	4	2	2	4	3	4	1	2	4	3	4	4	1	3	1	3
Que.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	3	1	2	1	2	1	1	4	2	3	4	3	4	2	3	4	2	3	2	2
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140
Ans.	2	1	4	4	4	4	3	3	2	2	1	1	3	1	4	3	4	2	4	3
Que.	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160
Ans.	2	3	4	3	2	1	3	2	4	2	4	4	2	2	4	4	1	3	3	4
Que.	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	3	1	1	4	3	1	2	4	1	3	3	1	2	2	1	2	3	3	3	4

HINT - SHEET

1. $-5\sin\theta + 12\cos\theta$

Maximum value $= \sqrt{(-5)^2 + (12)^2} = 13$

4. Unit of $\frac{C}{\lambda} = L^0 T^{-1}$

Unit of $\frac{x}{\lambda} = L^0$

Both have different dimensions

5. Area under a-t curve shows change in velocity so

$u = 0$

$V = \frac{1}{2}(10) \times 11 = 55 \text{ m/s}$

7. Range = 2 height

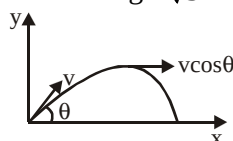
$$\frac{2u^2 \sin\theta \cos\theta}{g} = \frac{2u^2 \sin^2\theta}{2g}$$

$\tan\theta = 2$

so $R = \frac{2u^2 \sin\theta \cos\theta}{g}$

$$= \frac{2u^2}{g} \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{4u^2}{5g}$$

8.



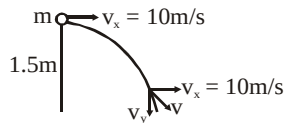
$\mathbf{v}_i = v \cos\theta \hat{i} + v \sin\theta \hat{j}$

$\mathbf{v}_f = v \cos\theta \hat{i}$

Change $\mathbf{v}_f - \mathbf{v}_i = v \sin\theta \hat{j}$

10. Body may have uniform velocity.
12. Earth has centripetal acceleration, as it revolves around sun hence not an inertial frame.
13. Speed of cart after 5 sec

$$v_x = (5)(2) = 10 \text{ m/s}$$



Now vertical component of speed after 0.5 sec

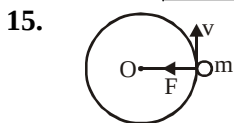
$$v_y = (0.5) \times 10 = 5 \text{ m/s}$$

so $v = \sqrt{v_x^2 + v_y^2}$

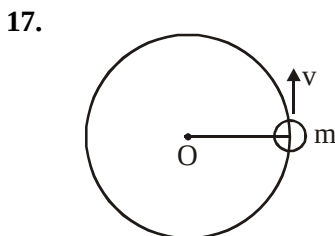
$$v = \sqrt{(10)^2 + (5)^2}$$

$$v = \sqrt{125}$$

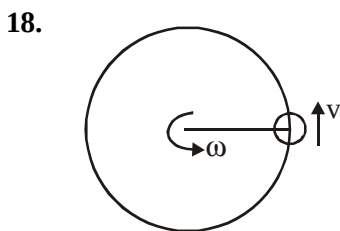
$$v = 5\sqrt{5} \text{ m/s}$$



as $\vec{v} \perp \vec{F}$, displacement is perpendicular to force applied hence $W = \vec{F} \cdot \vec{S} = 0$



In uniform circular motion only speed of particle is constant.



Here,

$\vec{\omega}$ & $\vec{\alpha}$ are axial vectors which cannot be perpendicular, angle between them be either 0° or 180° .

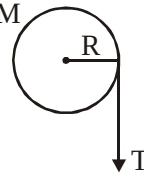
20. By Momentum conservation

$$\frac{50}{1000} \times 30 = M_{\text{gun}} \times 1$$

$$M_{\text{gun}} = \frac{50 \times 30}{1000} = 1.5 \text{ kg}$$

21. Centre of mass of a system of particle does not depend on forces acting on the particles.

23. M α - angular acceleration



$$TR = \frac{MR^2}{2} \alpha$$

$$\alpha = \frac{2T}{MR}$$

25. $\text{K.E.} = \frac{1}{2} I \omega^2$

Q $I = \frac{2}{5} mR^2$

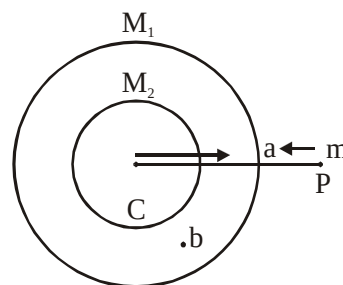
$$\text{K.E.} = \frac{1}{2} \cdot \frac{2}{5} mR^2 \omega^2$$

$$= \frac{1}{5} mR^2 \omega^2$$

$$= \frac{1}{5} \times 1 \times \frac{3}{100} \times \frac{3}{100} \times 50 \times 50$$

$$= \frac{9}{20} \text{ J}$$

- 26.



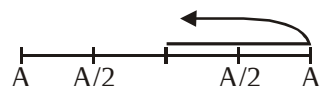
$$\text{Field at P} = \frac{G(M_1 + M_2)}{a^2}$$

$$\text{Force on } m = \frac{Gm(M_1 + M_2)}{a^2}$$

27. When particle at "b" field = $\frac{GM_2}{b^2}$

$$\text{force on 'm'} = \frac{GM_2 m}{b^2}$$

29.



$$\alpha = A \sin \omega t$$

$$A/2 = A \sin \omega t \text{ from mean position to } A/2$$

$$\omega t = \pi/6$$

$$t = \frac{\pi \times T}{6(2\pi)} = T/12$$

Now Time taken by body from mean to extreme
= $T/4$

So, From $A/2$ to A time taken will be

$$\frac{T}{4} - \frac{T}{12}$$

$$= \frac{2T}{12} = T/6$$

33.

$$\frac{1}{2l} \sqrt{\frac{T+\Delta T}{\mu}} - \frac{1}{2l} \sqrt{\frac{T}{\mu}} = 6$$

$$\frac{1}{2l} \sqrt{\frac{T}{\mu}} = 600$$

$$\frac{1}{2l} \sqrt{\frac{T}{\mu}} \left[\left(1 + \frac{\Delta T}{T} \right)^{1/2} - 1 \right] = 6$$

$$1 + \frac{\Delta T}{2T} - 1 = \frac{6}{600}$$

$$\frac{\Delta T}{T} = \frac{2}{100} = 0.02$$

37. P-N junction

38. Doping in base is less than emitter & collector.

39. A_v is highest in C.E.

$$\left[A_v = \beta \frac{R_0}{R_{in}} \right]$$

$$41. \Delta Q = ms\Delta T$$

$$\Delta Q = 0 \quad \therefore \quad S = 0$$

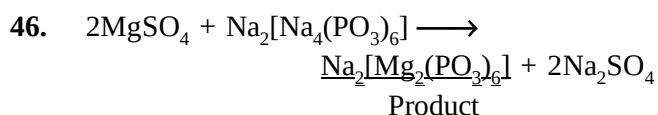
$$43. f = 5$$

$$K.E. = \frac{5}{2} RT$$

$$44. B = m(g-a)$$

$$a = g$$

$$B = 0$$

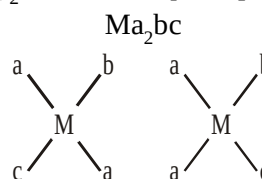


47.

$$\text{Thermal stability} \propto \frac{1}{\text{Polarisation}}$$

$$\text{Polarisation} \propto \frac{1}{\text{Size}}$$

48. $[Pd(NH_3)_2Cl Br] \rightarrow$ Square planar



49.

Co^{+3}	Cr^{+3}
$NH_3 : NO_2^-$	NH_3, NO_2^-
6 : 0	0 : 6
5 : 1	1 : 5
4 : 2	2 : 4
3 : 3	3 : 3
2 : 4	4 : 2
1 : 5	5 : 1
0 : 6	6 : 0

neutral not possible

50.

Sc^{+3}	$3d^0$	} No d-d transition
Ti^{+4}	$3d^0$	
Zn^{+2}	$3d^{10}$	
Cu^{+3}	$3d^3$	

d-d transition so colour

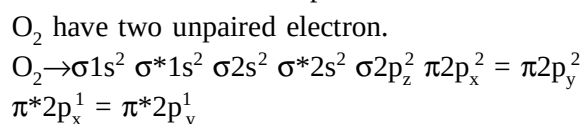
51.



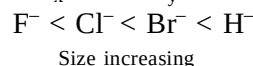
$$2\sigma \text{ bond} + 3 \text{ I.P.} = 5$$

$$= sp^3d$$

52.



53.

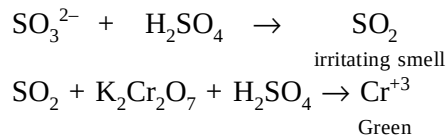


54.

$$I.P. \propto \frac{1}{\text{Size}}$$

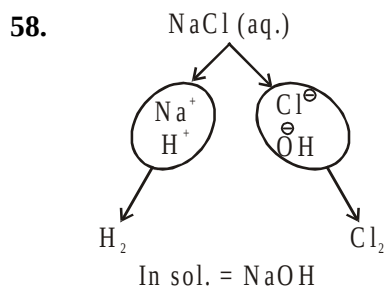
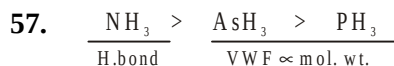
Size of He is minimum so I.P. of He maximum.

55.

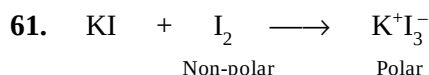
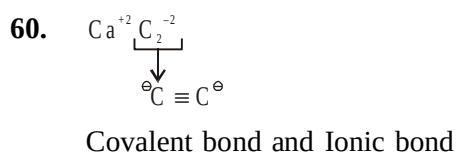


56.

F_2 is most oxidising agent that's why.

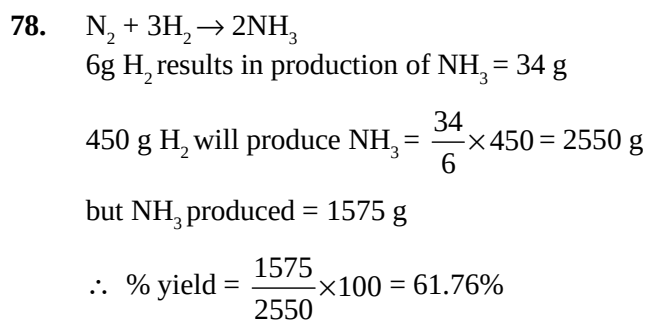
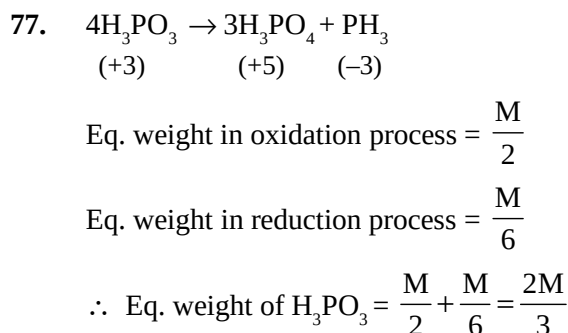


59. Leaching is only for Ag, Au, Al



76. I.P. = $E_\infty - E_1 = 36 \text{ V}$
 $\therefore E_1 = -36 \text{ V}$

second excitation potential = $E_3 - E_1 = -\frac{36}{9} - (-36)$
 $= 32 \text{ V}$



79. $r \propto \frac{1}{\sqrt{M}}$ since $r_{\text{unknown}} = 4 r_{\text{O}_2}$

$\frac{r_{\text{O}_2}}{r_{\text{unknown}}} = \sqrt{\frac{M_{\text{unknown}}}{32}}$

$\Rightarrow \frac{1}{4} = \sqrt{\frac{M_{\text{unknown}}}{32}} ; M_{\text{unknown}} = 2$

80. $d = \frac{Z \times M}{N_A \times a^3}$

$\therefore a^3 = \frac{1 \times 250}{6.02 \times 10^{23} \times 7.2} = 5.76 \times 10^{-23} \text{ cm}^3$

and $a = 3.86 \times 10^{-8} \text{ cm}$.

in simple cubic lattice $a = 2r$

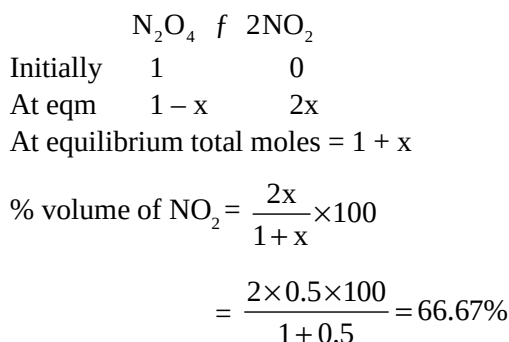
$\therefore r = \frac{3.86 \times 10^{-8}}{2} = 1.93 \times 10^{-8} \text{ cm}$.

81. $\Delta T_f = \frac{i \times k_f \times 1000w}{mW}$

$\therefore i = \frac{2 \times 172 \times 50}{1.72 \times 1000 \times 20} = 0.5$

83. $Q \frac{D}{d} = 1.5 = 1 + x$

$\therefore x = 0.5$



84. $q = 200 \text{ J}$

$W = -P\Delta V = -2 \times 10^5 \times 500 \times 10^{-6}$
 $= -100 \text{ J}$

$\Delta E = q + W = 200 - 100 = 100 \text{ J}$

86. $\text{pH} = \text{pKa} + \log \frac{[\text{salt}]}{[\text{Acid}]}$

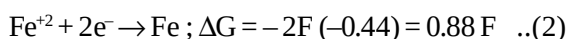
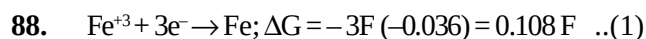
$\text{pH} - \text{pKa} = 5 = \log \frac{[\text{salt}]}{[\text{Acid}]}$

$\frac{[\text{salt}]}{[\text{Acid}]} = 10^5 ; [\text{salt}] = [\text{conjugate base}] = [\text{acid}] \times 10^5$

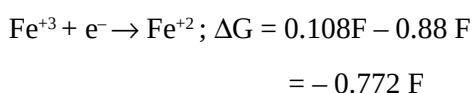
87. $t = \frac{2.303}{K} \log \frac{a}{a-x}$

$$t = \frac{2.303}{2.303 \times 10^{-3}} \log \frac{1}{0.25} = 10^3 \log 4$$

$$= 10^3 \times 0.6 = 600 \text{ sec.}$$

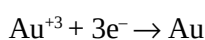
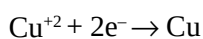
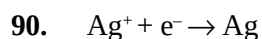


on subtracting equation (2) from equation (1)



and $E^\circ = -\frac{\Delta G}{nF} = -\frac{(-0.772F)}{1 \times F} = 0.772 V$

89. Sb_2S_3 is a negative colloid therefore Al^{+3} ion will behave as most effective coagulating agent.



deposited Ag = 3 mol

deposited Cu = $\frac{3}{2}$ mol

deposited Au = 1 mol

$\therefore$ molar ratio of Ag : Cu : Au = 6 : 3 : 2

93. NCERT (XII) Pg#140 para 7.9, (H) 151 para 7.9

94. NCERT XII Pg.# 149

97. NCERT XII Pg.# 159

98. NCERT (XI) Pg. # 168-169

99. NCERT (XI) Pg. # 166

100. NCERT (XI) Pg. # 139

111. NCERT (XII) Pg# 198, 199, 205

113. NCERT (XII) Pg# 182

115. NCERT (XII) Pg# 199

116. NCERT (XII) Pg# 205

117. NCERT (XII) Pg# 204

119. NCERT XIIth Pg.#99

120. NCERT XIIth Pg.#100

121. NCERT XIIth Pg.#104

122. NCERT XIIth Pg.#106

123. NCERT XIIth Pg.#110 Fig. 6.11

128. NCERT XIth Pg No. # 51

132. NCERT-XI Pg. # 101 para-1

133. NCERT-XI Pg. # 103

134. NCERT-XI Pg. # 120 para-4

135. NCERT XIIth Pg.# 5 Fig.1.2

138. NCERT Pg # 67 (Anatomy)

139. NCERT Pg # 80 (Morphology)

143. NCERT Pg. # 60

145. NCERT Pg. # 40

147. NCERT XIth Pg.#14

149. NCERT XIth Pg.#5

150. NCERT XIth Pg.#9

152. NCERT XIth Pg.#20

153. NCERT XIth Pg.#20

157. NCERT XIIth Pg # 303, para - 1

158. NCERT (XI) Pg. # 291, para - 2

159. NCERT XII pg.# 275

160. NCERT (XI) Pg. # 288,287

162. NCERT (XI) Pg. # 274, para - 1

174. NCERT (XI) Pg # 103(E), Para-3

179. NCERT (XI) Pg # 105, Para-2